

MODULE 4 – Control Flow & Decision Making

Module Objective

After this module, learners will:

- Make programs **decide and branch**
 - Understand **real-world logic**
 - Write **menu-driven & rule-based programs**
 - Be fully ready for **loops (Module 5)**
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4.1 What is Control Flow?

👉 Control flow decides **which code runs and which doesn't**.

Real life example:

4.2 The if Statement

python

 Copy code

```
age = 20

if age >= 18:
    print("Eligible to vote")
```

4.3 if-else Statement

python

 Copy code

```
age = 16

if age >= 18:
    print("Eligible to vote")
else:
    print("Not eligible")
```

4.4 if-elif-else (Multiple Conditions)

python

 Copy code

```
marks = int(input("Enter marks: "))

if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 60:
    print("Grade C")
else:
    print("Fail")
```

4.5 Nested if Statements

python

 Copy code

```
age = 25
has_id = True

if age >= 18:
    if has_id:
        print("Allowed entry")
    else:
        print("ID required")
else:
    print("Age restriction")
```

 Explain nesting with decision trees

4.6 Using Logical Operators in Conditions

python

 Copy code

```
age = 22
salary = 30000

if age >= 21 and salary >= 25000:
    print("Loan approved")
```

4.7 Ternary (One-Line if)

python

 Copy code

```
result = "Pass" if marks >= 40 else "Fail"
```

✓ Use for simple conditions only

4.8 match-case (Python 3.10+)

Why?

Cleaner alternative to long `if-elif`

python

 Copy code

```
day = int(input("Enter day number: "))

match day:
    case 1:
        print("Monday")
    case 2:
        print("Tuesday")
    case 3:
        print("Wednesday")
    case _:
        print("Invalid day")
```

💡 Similar to `switch-case` in other languages



4.10 Mini Programs (Practice)

✓ Program 1: Number Sign Checker

python

 Copy code

```
num = int(input("Enter number: "))

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

Program 2: ATM Menu

 Copy code

```
balance = 10000

print("1. Check Balance")
print("2. Withdraw")
choice = int(input("Enter choice: "))

if choice == 1:
    print("Balance:", balance)
elif choice == 2:
    amt = int(input("Enter amount: "))
    if amt <= balance:
        balance -= amt
        print("Withdraw successful")
    else:
        print("Insufficient balance")
else:
    print("Invalid choice")
```



4.11 Mini Project – Electricity Bill Calculator

Problem

Calculate electricity bill based on units:

Units	Rate
≤100	₹5/unit
101-200	₹7/unit
>200	₹10/unit

python

 Copy code

```
units = int(input("Enter units consumed: "))

if units <= 100:
    bill = units * 5
elif units <= 200:
    bill = (100 * 5) + (units - 100) * 7
else:
    bill = (100 * 5) + (100 * 7) + (units - 200) * 10

print("Total Bill: ₹", bill)
```